AEROSPACE SYSTEMS ENGINEERING PROPOSAL • PROBLEM 3 (TASKS 1–7 + BONUS)

Simulation vs. Practical Development for Competition RC Aircraft

0. Project Assumptions, Resource Constraints & Team Strategy

Baseline Assumptions: Our student team has a realistic **4-week window** and a modest budget of **₹25,000 (~$300 USD)** to build and fly a competition RC aircraft (b = 1.4 m, target mass $m_0 \approx 2.2$ kg). Available tools include basic workshop hand tools, a laser cutter, an entry-level 3D printer, and standard hobby electronics.
Chosen Approach — Balanced Hybrid Strategy: Rather than relying entirely on complex computer simulations (which often fail to account for real-world assembly imperfections) or rushing into blind trial-and-error building (which wastes materials and time when planes crash), our team adopts a balanced **hybrid approach**. We will use accessible software (XFLR5 / Fusion 360) for initial sizing and stability checks in Week 1, followed by straightforward bench testing and building a simple prototype to validate our design in reality.

1. CRITIQUE OF SIMULATION-LED VS. PROTOTYPE-LED DESIGN PARADIGMS

Task 1 & Task 7

Simulation-Led Design (XFLR5, OpenVSP, CAD)

- **Advantages:** Quick to test different wing shapes, airfoil sections, and static margin stability on a computer without spending money on materials. CAD helps ensure parts fit together nicely before cutting wood or foam.
- **Limitations:** Full CFD (like ANSYS) is too complex and time-consuming for student timelines. Simulations also miss real-world variables like glue weight, covering sag, servo slop, and rough grass runways [1].

Practical Prototype-Led Design (Build & Fly)

- **Advantages:** Gives immediate tactile feedback on structural strength, wiring layout, and how the plane actually handles in the air under real wind conditions [2].
- **Limitations:** Expensive and slow if a major design flaw causes a crash on the first flight. Rebuilding a whole wing from scratch takes days and depletes our limited budget.

2. QUANTITATIVE COMPARISON MATRIX ACROSS ENGINEERING METRICS

Task 2 & Task 7

Evaluation Metric	Pure Simulation-Led	Pure Prototype-Led	Balanced Hybrid Approach (Selected ★)
Budget / Cost	Low initially, but high risk if a simulated design crashes due to unverified build issues.	High consumable costs due to multiple crashed airframes and broken parts.	**Controlled (₹22,000)**: Free software checks + 1 simple test model + 1 final competition plane.
Development Time	Heavy focus on computer models; leaves very little time for actual flight testing.	Rushed building with little planning, leading to repeated trial-and-error fixes.	**Balanced:** 1 week digital planning, 1 week building test model, 2 weeks flight tuning.
Accuracy	Good for basic lift/drag estimates, but poor for predicting stall behavior and real weights.	High real-world fidelity, but hard to diagnose why failures happened without data.	**Reliable:** Digital baselines checked and corrected using simple flight tests.
Iteration Speed	Fast on computer (<5 mins per wing tweak), but slow if meshing errors occur.	Slow (takes 2–3 days to cut and glue a new wing after a structural failure).	**Moderate:** Quick digital checks for major dimensions; fast field adjustments for trim.
Failure Detection	Catches basic stability issues, but misses wiring failures or motor vibrations.	Catches physical issues, usually only after a hard landing or minor crash.	**Proactive:** Simple bench tests for motor thrust and wing strength before flying.
Final Performance	**Looks great on paper, but can be tricky to trim and fly on competition day.**	**Flyable and tough, but usually heavier and less aerodynamically efficient.**	**Optimized: Good balance of aerodynamic efficiency, structural strength, and reliability.**

Software Selection (Task 7): We will use **XFLR5** for quick 2D airfoil polars and 3D stability checks, **Fusion 360**

3. INTEGRATED CLOSED-LOOP DEVELOPMENT WORKFLOW

Task 3

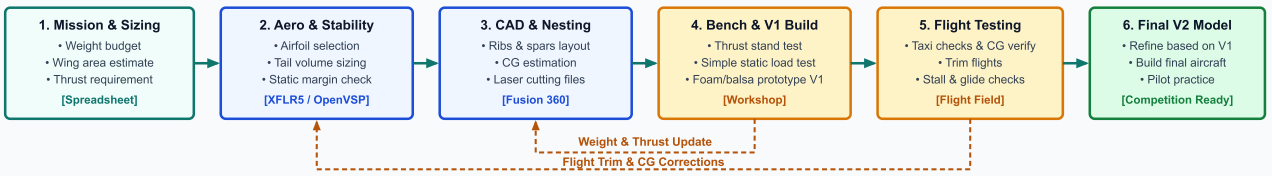


Figure 1: Closed-loop development workflow showing how initial digital design links directly to prototype testing and flight feedback.

Workflow Explanation for Defense: Our workflow is designed to be practical. Stage 1 and 2 establish basic sizing and stability on the computer. Stage 3 generates parts for laser cutting. Stage 4 involves simple bench tests (like checking motor thrust with a wattmeter) before building a test prototype. Stage 5 involves careful flight testing to find any trim issues, which are then fixed in Stage 6 for the final competition aircraft.

4. PARAMETER ALLOCATION: DIGITAL SIMULATION VS. PHYSICAL VERIFICATION

Task 4

Three Parameters Studied via Simulation (XFLR5 / CAD)

- **1. Longitudinal Static Margin & CG Range:** Checked in XFLR5 to ensure the center of gravity is safely ahead of the neutral point for stable pitch control.
- **2. Wing Lift Distribution & Twist:** Evaluated in XFLR5 to ensure the aircraft stalls gently at the wing root first rather than dropping a wingtip abruptly.
- **3. Component Layout & Weight Balance:** Modeled in Fusion 360 so we can place the battery and electronics correctly to hit our target CG without adding dead lead weight.

Three Parameters Verified Physically (Bench & Field)

- **1. Motor Thrust & Battery Voltage Sag:** Measured directly on a simple test stand using a digital scale and wattmeter to verify real-world thrust.
- **2. Wing Structural Strength:** Tested physically by supporting the built wing and placing weighted bags on it to ensure it easily handles expected flight loads.
- **3. Actual CG and Control Surface Throws:** Checked physically on a balancing stand and measured with a protractor once the complete aircraft is built and covered.

5. METHODOLOGY FOR VALIDATING SIMULATIONS USING MEASUREMENTS

Task 5

To ensure our computer models match reality, we use straightforward, low-cost validation checks [3]:

- **1. Simple Glide Tests (Checking Sink Rate & Trim):** During calm morning test flights with the motor off, we check how smoothly the aircraft glides. If it sinks faster than expected in simulation, we know our real-world drag is higher due to film seams and pushrods, helping us refine our expectations.
- **2. Stall Speed Verification:** We perform gentle, high-altitude stalls to see at what speed the nose drops. Comparing this real stall speed to our XFLR5 calculations tells us if our estimated weight or maximum lift coefficient was accurate.
- **3. Static Thrust & Current Check:** We compare our simulated motor performance against physical wattmeter readings on the bench to ensure our ESC and battery won't overheat during full-throttle takeoffs.

6. FAILURE ANALYSIS & DIAGNOSTIC PLAN: SIMULATED PLANE FLIES, PROTOTYPE CANNOT TAKE OFF

Task 6

Understanding the Problem: If an aircraft performs well in simulation but refuses to take off in reality, it means real-world forces (like rolling friction in grass, thrust loss, or incorrect ground angle) are overpowering the plane [1], [4]. Here are the 6 most common causes and our step-by-step diagnostic plan:

Issue Category	Plausible Real-World Cause (Not Captured in Sim)	Field Observation / Symptom	Diagnostic Check & Fix
1. Landing Gear & Ground Angle	Main wheels placed too far back, or nose strut too short, pointing the wing downward into the ground during takeoff roll.	Plane rolls fast down the runway, but pulling up on the elevator fails to lift the nose.	**Check:** Measure ground angle with a digital level. Lengthen nose gear slightly to give positive wing incidence.
2. Extra Weight & Forward CG	Glue, wiring, and paint added unexpected weight, and the CG ended up too far forward, limiting elevator authority.	Plane feels sluggish on takeoff or lifts briefly then settles back down onto the runway.	**Check:** Weigh the finished plane and re-check balance on the CG stand; shift battery back if needed.
3. Low Motor Thrust	Propeller mounted backward, or battery voltage sagging under load due to high internal resistance.	Distinct lower propeller sound; plane reaches a low terminal speed and refuses to accelerate further.	**Check:** Verify propeller text faces forward; check throttle calibration on the radio and test with a wattmeter.
4. Control Linkage Flex	Pushrods bending under aerodynamic loads, meaning the elevator isn't moving as much in flight as it does on the bench.	Control surfaces move fine on the ground, but become ineffective at higher speeds.	**Check:** Support the elevator by hand while moving the servo to check for pushrod flex; add support guides.
5. High Runway Friction	Flying off thick grass with small wheels that sink into the turf or bind against their axles.	Plane accelerates poorly and slows down immediately when throttle is reduced.	**Check:** Lubricate wheel axles, check for wheel binding, and upgrade to slightly larger lightweight wheels if needed.
6. Covering Sag / Wing Twist	Iron-on covering film sagged between ribs in the sun, ruining the airfoil shape and lifting performance.	Plane reaches takeoff speed and pops off the ground, but immediately stalls or drops a wing.	**Check:** Inspect wing surface for sagging; re-tighten covering film with an iron and check wing alignment.

20-Minute Field Diagnostic Protocol: If takeoff fails, our team will follow a quick checklist: (1) Check weight and CG balance, (2) Test full-power motor thrust and check propeller direction, (3) Inspect control throws and pushrod stiffness, and (4) Check ground angle and wheel roll freedom.

7. BONUS (+5): REALISTIC FOUR-WEEK SCHEDULE WITH CONTINGENCY BUFFERS

Bonus Section

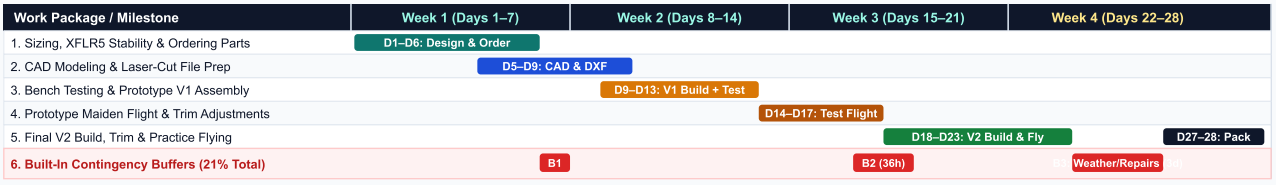


Figure 2: Four-week schedule featuring 6 days of explicit contingency buffers to absorb shipping delays, weather holds, or minor repairs.

Contingency Rationale: We have reserved 6 days (21.4% of the schedule) across three buffers (B1 for parts shipping, B2 for test flight adjustments, and B3 for unexpected repairs right before competition) so our team isn't rushing under panic.